

Hints

Vayu Air · Session 3: one nudge per question

How to use these

Tip Use a hint only after you have tried the question. Each hint points you at the right idea, not the answer.

Hints by question

Question 1. Write the grain as a comment first, the way the session showed. One booking is one ticket on one flight. Everything describing the who, what, where and when is a dimension key; the numbers you would SUM are measures.

Question 2. Dimensions get an integer surrogate primary key (an IDENTITY column works well) and keep their original id as a business key. The fact holds the foreign keys plus the additive measures, at the grain from Question 1.

Question 3. Build DateKey as `YEAR(d)*10000 + MONTH(d)*100 + DAY(d)`. To load the fact, join `bronze_bookings` to each dimension on the business key and select the dimension's surrogate key.

Question 4. Split airport into airport, city and country tables, each referencing the next with a foreign key. Load country first, then city, then airport, so the foreign keys resolve. The trade-off is fewer repeated values versus more joins.

Question 5. This is the two-pass Type 2 pattern from the session. Step one: a MERGE that, when the tier or home airport differs, sets `is_current = 0` and `effective_to`, and inserts brand-new passengers. Step two: a separate INSERT that adds the new current version for everyone you just expired.

Question 6. Use `CREATE PARTITION FUNCTION ... AS RANGE RIGHT FOR VALUES (...)` with monthly DateKey boundaries, then `CREATE PARTITION SCHEME`, then put the fact ON `the_scheme(DateKey)`. Filtering on DateKey prunes; filtering on another column cannot. Check "Actual Partition Count" in the execution plan.

Question 7. Bronze is the raw landed source, silver is cleaned and conformed with the SCD2 dimension, gold is the partitioned star that feeds dashboards. For the contract, think schema and types, allowed values, freshness, ownership, and how a change is communicated. A new optional column is non-breaking; renaming or removing a column is breaking.